

Secure Circuits

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PRECISION. PERFORMANCE. POSSIBILITIES.

PRINTED CIRCUIT BOARD

Powering Innovation. Connecting the Future.



HIGH PRECISION

Advanced technology for superior accuracy



RELIABLE & DURABLE

Built to last with high quality components



100% TESTED

Rigorous testing for defect-free performance



FAST DELIVERY

On-time delivery, every time



HIGH
PERFORMANCE



ADVANCED
TECHNOLOGY



ROHS
COMPLIANT



QUALITY
ASSURED



CUSTOM
SOLUTIONS



INNOVATIVE SOLUTIONS
FOR A SMARTER WORLD

Company Profile

- Established in 2020, we are PCB design, manufacturing, and assembly experts.
- We serve industries like Automotive, Industrial, Power, Communication, Aerospace, and Energy Metering.
- Our in-house facility supports fast prototyping and mass production of single and double-sided PCBs with high quality and on-time delivery.



Mr. Mukesh K Purvey

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Founder & CEO

Mr. Mukesh K Purvey is a seasoned PCB industry professional with 20+ years of experience in design, CAM, manufacturing, assembly, and R&D. He holds a B.E. in Electrical Engineering from MS University Pune. His expertise and leadership have driven strong company growth. He is actively associated with IPCA and other industry bodies, keeping the company aligned with the latest technological advancements.

1. PROCESS FLOW CHART FOR SINGLE SIDE PCB



1. PROCESS FLOW CHART FOR DOUBLE SIDE PCB



Sales & Marketing

- **Build strong customer relationships through regular communication and support.**
- **Gather and implement customer feedback to improve service quality and satisfaction.**
- **Analyze Gerber data for accurate and competitive quotations.**
- **Perform preliminary design checks to ensure manufacturability.**
- **Collaborate closely with customers to meet technical and commercial requirements.**
- **Focus on developing long-term business partnerships and new growth opportunities.**

CAD-CAM Engineering

- **Critical process in PCB manufacturing ensuring design accuracy and manufacturability.**
- **Experienced CAD-CAM engineers collaborate closely with production teams throughout the product lifecycle.**
- **Expertise in handling complex and high-precision PCB designs.**
- **Focus on optimizing designs for cost-effective and efficient manufacturing.**
- **Commitment to high quality, reliability, and customer satisfaction.**

Material Cutting

Sheet Cutting & Job Launching

- Ensures proper material selection based on project requirements.
- Verifies material specifications before production initiation.
- Optimizes panel utilization to minimize material wastage.
- Conducts design and manufacturing readiness checks.
- Uses high-quality materials to ensure product reliability and performance.
- Establishes a strong foundation for efficient and accurate PCB manufacturing.



Drilling

High-Quality PCB Assembly & Manufacturing

1. Engineering Review

- Verification of Gerber data and customer specifications.
- Design optimization for manufacturability

2. Material Preparation

- Selection of high-quality laminates, copper-clad materials, and consumables.
- Material inspection to ensure compliance with specifications.

3. PCB Fabrication

- Precision manufacturing using advanced production equipment.
- Strict process controls to maintain dimensional accuracy and quality.

4. Assembly Process

- Accurate component placement and soldering.
- Support for both prototype and volume production requirements.

5. Quality Inspection

- Comprehensive inspections at every stage of production.
- Electrical testing and quality verification to ensure defect-free products.

6. Final Packing & Dispatch

- Secure packaging to prevent transit damage.
- Timely dispatch with complete documentation and traceability.

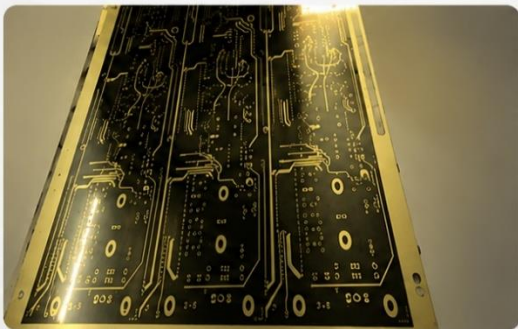


Image Processing

- Converts customer design data into production-ready manufacturing files.
- Performs image generation and artwork processing with high accuracy.
- Verifies PCB layouts, layer alignment, and design integrity.
- Optimizes artwork to ensure precise circuit pattern transfer.
- Conducts quality checks to eliminate design and imaging errors.
- Supports complex, high-density, and multilayer PCB designs.
- Utilizes advanced software and imaging technologies for maximum precision.
- Ensures compliance with customer specifications and industry standards.
- Enhances manufacturing efficiency, consistency, and product reliability.
- Focuses on delivering high-quality PCB solutions with accuracy and on-time execution.



Circuit



Track Exposing panel



Panel after Track Developing.

Final Inspection & Dispatch

- Comprehensive microscope inspection is performed to verify PCB quality and workmanship.
- Critical parameters such as track integrity, hole quality, solder mask, and surface finish are thoroughly checked.
- Any defects or non-conformities are identified and rectified before approval.
- Only PCBs that meet all quality and customer specifications are cleared for shipment.
- Final quality approval ensures reliability, performance, and compliance with industry standards.
- Approved PCBs are securely packed and prepared for timely dispatch to customers.



Stage One – Circuit Exposure & Development

- Verification of proper circuit exposure and image transfer accuracy.
- Inspection of developed circuit patterns for clarity and completeness.
- Ensures tracks, pads, and design features match the approved artwork.
- Detects any exposure or development defects before further processing.
- Maintains high precision and quality standards in PCB manufacturing.



Stage Two – Etching Inspection

- Verifies complete removal of excess copper during the etching process.
- Inspects PCB tracks for cuts, opens, and short circuits.
- Checks for under-etching or over-etching defects.
- Ensures circuit dimensions and conductor widths meet design specifications.
- Confirms accurate pattern transfer and electrical continuity.
- Maintains high-quality standards before proceeding to the next manufacturing stage.



Stage Three – Final Surface Finish & Legend Inspection

- Verifies proper **Solder Mask**, **HASL finish**, and **Legend Printing** quality.
- Ensures clear and accurate component markings and identification.
- Checks for blocked or incomplete plated holes.
- Inspects PCB tracks for cuts, opens, and short circuits.
- Confirms surface finish quality and overall board appearance.
- Ensures compliance with customer specifications and quality standards before final approval.



6. Etching

- Removes unwanted copper to create the required circuit pattern.
- Ensures accurate track width and circuit definition.
- Critical process for achieving high-quality PCB circuitry.

Etchants Used

1. Ammonium Persulfate

- Fast and efficient etching process.
- Ideal for fine-line and high-precision circuits.
- Produces clean and accurate copper patterns.

2. Cupric Chloride

- Controlled and uniform etching rate.
- Provides excellent edge definition.
- Minimizes undercutting for improved circuit accuracy.

Benefits

- High precision circuit formation.
- Consistent etching quality.
- Reliable electrical performance and manufacturability.

- The plated laminate panel undergoes a controlled etching process to remove unwanted copper.
- A photoresist mask is applied to protect the required circuit traces and drilled holes.
- Exposed copper areas are chemically etched away using specialized etchants.
- The protected copper remains intact, forming the desired circuit pattern.
- Ensures accurate track geometry, circuit definition, and dimensional precision.
- Critical for achieving reliable electrical performance and high-quality PCB manufacturing.
- Final inspection is carried out to verify etching quality and circuit integrity before proceeding to the next process stage.



Printing Process

- PCB panels undergo **QC2 inspection** before entering the printing stage.
- A **Solder Mask** is applied to protect copper circuits and prevent solder bridging.
- Individual PCB boards are properly defined and isolated during the masking process.
- **Legend Printing (Silkscreen)** is carried out to print component markings, part numbers, logos, and identification details.
- Different colored masks and inks are applied as per customer and product requirements.
- Printing quality is verified for clarity, alignment, and accuracy.
- After completion of masking and legend printing, the panels undergo **QC3 inspection**.



Component Marking for Efficient Assembly & Debugging

- Clear component markings ensure accurate and efficient PCB assembly.
- Labels such as **R (Resistor)**, **C (Capacitor)**, and **U (IC)** enable quick component identification.
- Improves assembly speed and placement accuracy.
- Minimizes the risk of component placement errors.
- Simplifies inspection, testing, and troubleshooting processes.
- Enhances overall manufacturing quality and reliability.



HASL (Solder Coating) & Solder Mask Key Points

- **Surface Finish Purpose:** Protects exposed copper traces from oxidation and corrosion, significantly increasing PCB durability.
- **Assembly & Solderability:** Enhances the board's solderability, directly impacting overall assembly quality and long-term reliability.
- **Solder Mask Protection:** Acts as a protective layer over copper traces to prevent oxidation and accidental short circuits.
- **Solder Paste Definition:** Strictly defines the specific areas where solder paste should be applied, ensuring accurate component placement and reliable soldering joints.



Final Testing with FPT (Flying Probe Testing)

- **Equipment Standard:** Secure Circuits utilizes advanced, state-of-the-art testing systems to ensure bare board precision and electrical integrity.
- **Mania Speedy & ESL-608:** High-precision flying probe machines deployed for fixtureless electrical testing, verifying circuit continuity, and isolating defects across complex PCB designs.
- **Bare Board Tester (BBT):** Serves as the final quality checkpoint to test unpopulated PCBs for open circuits or structural shorts before shipment.
- **High-Speed Performance:** Capable of executing rapid electrical testing at speeds of 3,000 to 3,200 points per minute.
- **Adjustable Probe Pressure:** Ranging from 5 to 25 grams, allowing safe and efficient testing of fragile or delicate boards.
- **Micro-Precise Positioning:** Achieves an ultra-accurate probe positioning tolerance of $\pm 0.01\text{mm}$.
- **Advanced CCD Calibration:** Employs high-definition CCD camera technology for superior alignment and calibration over traditional contact methods.



Sec

Major PCB Manufacturing Machines

Process	Machine Used
Cutting	PCB Shearing Machine / Router
Drilling	CNC Drilling Machine
Plating	Copper Plating Line
Exposure	UV Exposure Machine
Etching	Etching Machine
Solder Mask	Screen Printing / Coating Machine
Testing	Flying Probe Tester
Inspection	AOI Machine

PCB manufacturing is a highly controlled process involving precision machinery, chemical treatment, imaging systems, and electrical testing to ensure high-quality boards.